

Address: Bharat Colony, Faridabad, Haryana

LinkedIn: linkedin.com/in/aman-sharma-7498501b9

Portfolio: 20lucifer.github.io/Portfolio

SUMMARY OF QUALIFICATIONS

Embedded Firmware and Edge AI Engineer with three years in firmware development and two years on embedded Linux platforms. Develops and optimizes firmware for microcontroller-based systems, specializing in Vehicle Tracking Systems (VTS) and AIS-140 compliance, and deploys **computer vision** and **object detection models** on edge hardware such as **i.MX93 FRDM**, **custom i.MX95 (15×15 and 19×19)** and STM32MP2 platforms. Proficient in Embedded C, **Python**, communication protocols, flash memory management, bootloaders, hardware-software integration, **TensorFlow Lite**, **ONNX**, **INT8 quantization** and **NPU-accelerated inference**. Strong background in debugging, performance optimization and mentoring junior engineers, and adept at collaborating with cross-functional teams.

PROFESSIONAL EXPERIENCE

Embedded Design Services India Pvt. Ltd.**Faridabad, Haryana****Oct 2022 – Present**

Embedded Linux AI Developer: (Joined as an Intern in 2022, promoted to Sr. Firmware Design Engineer and then to Embedded Linux AI Developer.) Trained and deployed **object detection models (YOLO, SSD, MobileNet, EfficientNet)** for **real-time vision** applications on embedded Linux platforms. Converted and quantized models to **TensorFlow Lite** and **ONNX** using **INT8 post-training quantization**, and accelerated inference on **NPU**s with **Arm Ethos-U/Vela**, **NXP eIQ Neutron** and **ST AI hardware**. Built camera and video pipelines using **OpenCV**, **GStreamer**, **V4L2** and **MIPI CSI** on **i.MX93 FRDM**, **custom i.MX95 (15×15 and 19×19)** and **STM32MP2** boards.

Sr. Firmware Design Engineer: Developed and optimized embedded firmware for microcontroller-based Vehicle Tracking Systems (VTS), ensuring compliance with AIS-140 and AIS-038 standards. Designed modular libraries and firmware around UART, SPI, I2C and CAN interfaces to improve code reusability and maintainability. Implemented and optimized communication over TCP/IP, HTTP and MQTT protocols. Worked extensively on NAND flash memory management for efficient data logging. Developed a bootloader for firmware updates over-the-air (OTA). Integrated BLE modules for diagnostics and dynamic configuration. Conducted hardware-software integration and debugging using oscilloscopes and logic analyzers.

Leadership & Collaboration: Mentored junior engineers in embedded development and debugging techniques. Collaborated with cross-functional teams including hardware, testing and project management. Conducted peer code reviews and ensured adherence to embedded coding standards.

KEY PROJECTS

(i) Driver Monitoring System (DMS) – Edge AI: Built a **real-time driver monitoring pipeline** that detects **phone usage and smoking behavior** on **edge hardware**. Extracted driver behavior alerts and images from the DMS and transmitted events to monitoring systems over TCP/HTTP.

(ii) Railway Line-Crossing Gate Automation: Developed a **computer vision pipeline** that recognizes **approaching trains and vehicles** to automate level-crossing gate control.

(iii) Metro Yellow-Line Alert Detector: Implemented **real-time person detection** on platform camera feeds to identify passengers crossing the yellow safety line.

(iv) Security Surveillance Analytics: Deployed **on-device person and vehicle detection** for surveillance cameras, enabling **low-latency alerts without cloud dependency**.

(v) DMS + VTS Integration: Combined DMS with VTS to monitor driver behavior. Enabled real-time alert sending to TCP server and image upload to HTTP server.

(vi) Vehicle Tracking System (VTS) Projects: Developed firmware for AIS-140 and AIS-038 compliant VTS for clients like TML, IOCL, AEPL, Nippon and JIO. Implemented real-time alerts (ignition, over-speed, panic), GPS data logging and vehicle diagnostics. Used Quectel EC200 4G LTE module for communication (TCP, HTTP, MQTT), integrated BLE for local configuration and maintained a modular firmware architecture supporting multiple customer configurations and protocols.

(vii) RGBAW Stick with Wireless Control (RFM Remote + RFM Integration): Developed and maintained firmware for the Simpex and Prolight RGBAW Stick, including encoder logic and dynamic color control. Integrated an RFM wireless module and built a dedicated RFM-based remote control, optimizing signal integrity and response time for reliable remote-controlled lighting operation.

(viii) Low Speed Warning System (LSWS) for EV Buses: Developed a continuous virtual motor sound system that plays synthetic engine noise while an electric bus is in motion, enhancing pedestrian awareness at low speeds in line with EV acoustic vehicle alert requirements.

(ix) BLDC Motor Control: Worked on firmware-level feature additions and modifications for an existing BLDC motor control system, implementing new functional requirements and configuration updates based on limited access to source code, and developed driver code for BLDC regulators.

(x) Battery Management System, Battery Level Indicator & Load Dump Generator: Analyzed and maintained legacy BMS firmware and safety-critical battery power management logic during training. Implemented logic to measure and display battery levels accurately, and designed firmware to test PCBs for high-voltage surge (100V for 10ms) compliance.

EDUCATION, COMPETENCIES, LANGUAGES & INTERESTS

M.B.A. – Information Technology, Amity University, Noida 2023-2025

B.Tech. – Electronics & Communication Engineering, Satyug Darshan Institute of Engineering & Technology 2019-2023

Higher Secondary Education – Science, Govt. Model Sr. Secondary School 2017-2019

Embedded Skills: Embedded Firmware Development, Embedded C, STM32, ARM Cortex-M, IoT, DMA, Timer, ADC, OLED, LCD

Communication Protocols: USART/UART, SPI, I2C, CAN, TCP/IP, HTTP, MQTT, BLE

Edge AI & Vision: Python, OpenCV, YOLO, SSD, MobileNet, EfficientNet, Object Detection, Image Processing

Model Deployment: TensorFlow Lite, TFLite Runtime, ONNX, INT8 Quantization (PTQ), NPU Acceleration (Ethos-U, Vela, Neutron, ST AI)

Embedded Linux: Yocto, BitBake, BSP Development, Linux Kernel, Device Tree, GStreamer, V4L2, MIPI CSI

Platforms: STM32MP2, i.MX93 FRDM, Custom i.MX95 (15×15), Custom i.MX95 (19×19)

Domains: VTS (AIS-140), LED Lighting System (Simpex); **Tools:** IAR Embedded Workbench, Oscilloscope, Logic Analyzer

Professional Skills: Project Management, Code Review Practices

Soft Skills: Team Building/Collaboration, Effective Communication, Problem Solving, Critical Thinking

Languages: Hindi (First Language), English (Upper-Intermediate)

Hobbies & Interests: Playing Online Games, New Technologies, Watching Anime/Movies